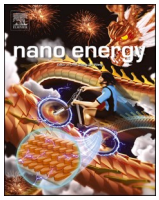
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High-loading homogeneous crosslinking enabled ultra-stable vertical organic electrochemical transistors for implantable neural interfaces

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ABSTRACT

Organic electrochemical transistors (OECTs) with high transconductance, low driving voltage, and biocompatibility have emerged as dominant bioelectronics technologies. However, a significant challenge remains in achieving stable signal recording over extended durations, due to the restricted thermal and cycling stability. Here, by introducing a small molecular with double-end functionalized trifluoromethyl-substituted diazirine (DtFDA) as the crosslinker in the organic mixed ionic-electronic conductors (OMIECs), both p- and n-type vertical OECT (vOECT) achieves excellent thermal adaptability (> 100 °C), excepti°nal cycling stability (> 200,000 cycles), fast ion transporting capability (transient time < 1 ms), along with ultrahigh maximum transconductance (g_m) (> 0.34 S). The findings reveal that highly loaded crosslinkers (OMIEC:DtFDA = 1:1 in mass) triggered by ultraviolet can facilitate the formation of stable and homogeneous transistor channel, which holds lower crystallinity, wider *d* spacing, and face-on dominated packing, thereby leading to enhanced ion hydration/injection and decent carrier transport pathways. Moreover, in vivo brain recordings with stable signals were accessed from the developed ultraflexible vOECT even after a four-week interval, and the tissue biocompatibility was also validated by chronic epicortical implantation. This work signifies new possibilities for high-performance vOECTs with high fidelity signals recording under complicated operation environments, enabling broader implications for robust bioelectronics.

1. Introduction

Since the first demonstration of organic electrochemical transistors (OECTs) in 1984 [1], tremendous research attentions have been attracted due to the superiorities of OECTs in low power consumption, high amplification capability, and biological compatibility, which matches well with the demands of next-generation electronics [2–4], including biosensors [5–7], wearable electronics [8–10], neuromorphic computing [11–16], human-machine interface [17], etc. The foundation of OECTs is the organic mixed ionic-electronic conductor (OMIEC) based channel, which simultaneously holds ionic and electronic conductivities. Specifically, ion injection/ejection dominates the doping/dedoping status of the OMIEC (typically polymers), leading to high or low carrier concentration. A decent OMIEC requires both high ion and electron/hole mobilities, thus the transistor channel can be doped/dedoped in a short time, leading to fast switching and effective modulation of the channel resistance [18–22]. Therefore, OMIECs typically hold hydrophilic side chains for facilitated hydration to achieve ion mobility. Previous research has revealed that hydration dynamics of OMIECs upon electrochemical bias differ dramatically between the amorphous and crystalline sites, where ion transport is facilitated and dominated within amorphous regions [23,24].

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Consequently, OECTs based on OMIECs with relatively low crystallinity show fast switching due to enhanced ion mobility, but the electron/hole charge transport pathway can be disrupted and show rapid decay due to uncontrollable swelling effect [25]. Accordingly, coexistence of crystalline and amorphous domains was concluded to be highly desirable and various strategies, including novel OMIEC synthesis and channel morphology engineering, have been applied for facilitated ion uptake and enhanced electrical properties of the OECT channel [26,27].

On the other hand, inevitable thermal effects during device storage/operation and uncontrollable electrolyte hydration still result in evident device performance degradation, due to unstable film microstructure, as the crystallinity tends to increase under elevated temperature along with reduced active content in OMIEC based channel after multiple swelling process [19,28]. In detail, thermal stress can lead to enriched crystallized sites, that would impede hydration process and ion injection/ejection pathways, which further slows down or even prohibits the doping/dedoping process of the channel. While swelling during multiple cycling process would lead to irreversible variation on the morphology of OMIEC channel, thus disrupt effective and stable ion and electron/-hole transport [29,30]. Consequently, addressing these obstacles are fundamental for developing OECTs with reliable performance for next-generation bioelectronics.

In the meantime, the introduction of an advanced vertical architecture enables OECTs with record high transconductance ($g_m > 0.1$ S) and cycling stability (> 50,000 cycles), which benefit from enhanced ionic-electronic coupling due to confined channel length down to ~100 nm, thus providing new possibilities for various applications [31–35]. While, in vertical OECTs (vOECTs), since the ions diffusion length (15 μm for vOECTs with 30 μm top electrode) can be > 100 times longer than that (typically ~ 100 nm) in a conventional planar OECT (cOECT), variation of ion mobility of OMIECs can lead to tremendous differences in device performance [31,36]. In fact, to obtain decent vOECT performance, a cooling stage that can avoid/minimize the unwanted thermal annealing of the OMIEC during the thermal evaporation of the Au top electrode, is mandatory, as the performances of vOECTs would degenerate rapidly after thermal effect during the evaporation of top electrode. On the other hand, a few researches have revealed cOECTs with relatively good thermal adaptivity by introducing solid state electrolytes, but the ion diffusion lengths in these transistors are limited (< 100 nm) [37–39]. Thereupon, to accelerate the development of vOECTs, and meet the stability demands (both thermal adaptivity and cycling stability) for various bioelectronic applications, vOECTs with good resistance to thermal stress, along with high g_m, fast and stable switching, are highly desired.

Here, we demonstrate p- and n-type vOECTs with decent thermal adaptivity, fast switching, and ultra-stable cycling stability via introducing a photo cross-linkable small molecular (double-end functionalized trifluoromethyl-substituted diazirine, DtFDA) in poly(2,5-bis(3-triethylene-glycoloxythiophen-2-yl)-co-thiophene) (P(g2T-T), p-type OMIEC) or polyethylene glycol substituted poly((N,N′-bis(2-octyldodecyl)-1,4,5,8-naphthalenedicarboximide-2,6-diyl)-alt-5,5′-(2,2′-bithiophene)) (PEG-N2200, n-type OMIEC), respectively. Especially, highly loaded cross-linker (OMIEC:DtFDA = 1:1 in mass) in the vOECT channel leads to balanced ionic and electronic charge transport, simultaneously, along with ultra-stable fabrication/operation fidelity. Thermally stable (up to 100 °C) channel microstructure promote the realization of vOECTs with high g_m (~340 mS), current on-off ratio (> 10^5), and excellent operational stability (over 200,000 cycles). Furthermore, we developed ultraflexible neural interface employing high-loading cross-linking strategy, such vOECT demonstrated remarkable neural signal stability during in vivo brain recordings, and long-term cortical epithelial membrane implantation biocompatibility. This work provides an easy and universal design strategy for high-performance vOECTs with decent fabrication and operation reliability, which could accelerate the further commercialization of relevant bioelectronics.

2. Results and discussion

The schematic illustration and working principle of the vOECT are illustrated in Fig. 1**a**, which relies on the gate potential-triggered lateral injection of ions that drift from the electrolyte into the semiconductor bulk through the edges of the top gold electrode, counterbalancing the electronic charges in the semiconductor. However, thermal effect incurred during device fabrication, operation, and storage precipitate a transition in OMIEC films from amorphous to crystalline states, thereby engendering substantial disparities in hydration kinetics when subjected to electrochemical bias, especially restricted ion transport and exchange over micrometre scale lengths (Fig. 1**b**). Accordingly, thermally resilient OMIEC films are highly desirable for efficient and stable mixed ion-electron conduction in vOECTs for bioelectronic applications. To meet this requirement, chemical structures of the materials used in this study are shown in Fig. S1, including two OMIECs (P(g2T-T) and PEG-N2200), DtFDA (cross-linker) [40,41] and cinnamate-cellulose (Cin-Cell) (encapsulation layer) [42]. The crosslinking mechanism of DtFDA as the cross linker of OECT channel is illustrated in Fig. 1**c**, where DtFDA holds trifluoromethyl-substituted diazirine group at both ends. Such group can release nitrogen gas and form a singlet nitrene upon UV (365 nm) activation, the singlet nitrene then attacks available sp^3 C-H bonds in the ethylene glycol (EG) side chains of OMIECs, and leads to covalent crosslinking [43–45]. In detail, the main photo-crosslinking products are derived from the insertion of non-conjugated C-H bonds by carbene reaction [40] and without disrupting the conjugation in the backbone, which still ensures efficient charge transport properties of the semiconducting polymer. Therefore, direct photopatterning of the P(g2T-T)/DtFDA blend film (6:1 in mass) could be accomplished by exposing the film under UV light (365 nm, 1500 mW) for 2 min with a shadow mask, developing in chloroform for 1 s, and blow drying, subsequently (Details of the fabrication/patterning process can be found in Experimental Section). Representative optical images of the P(g2T-T)/DtFDA blend films with different patterns are illustrated in Fig. 1**d**. It is obvious that, DtFDA acts as an efficient and precise photo crosslinker, which leads to patterns with resolution < 1 μm (Fig. S2).

Next, microstructures of these blend films (including neat P(g2T-T) film, and P(g2T-T)/DtFDA blend films with mass ratios of 6:1, 3:1, 1:1 and 1:3, respectively) were subsequently investigated by atomic force microscopy (AFM). AFM height images of the blend films (Fig. S3a) indicate that all films exhibit ultra-smooth surfaces with root-mean-square roughness (σ_{RMS}) of ≤ 0.71 nm, where increased DtFDA content would slightly decrease σ_{RMS} from 0.71 nm (neat P(g2T-T)) to 0.45 nm (P(g2T-T):DtFDA = 1:3 in mass). After being subjected to thermal stress (150 °C for 10 min on a hot plate in air), neat P(g2T-T) and blend film with high P(g2T-T) content (P(g2T-T):DtFDA = 6:1) films show obvious increased surface roughness σ_{RMS} from 0.71 nm and 0.67 nm to 1.19 nm and 1.13 nm, respectively (Fig. S3b), indicating possible microstructural reorganization due to thermal stress. While blend films with high DtFDA content (P(g2T-T):DtFDA = 3:1, 1:1 and 1:3) show minor enhanced σ_{RMS} from 0.59 nm, 0.55 nm and 0.45 nm to 0.72 nm, 0.63 nm and 0.52 nm, respectively, revealing stronger resistance to thermal stress. Moreover, from the AFM phase images, with high DtFDA content, no significant phase change before and after thermal stress are observed. While films with low/no DtFDA show distinct grain features, implying possible transformation from amorphous state to more crystalline state.

Furthermore, through the characterization of UV–vis spectroscopy (Fig. S4), it is observed that in all films, the relative intensity of 0–0 (at ~ 662 nm) and 0–1 (at ~ 614 nm) absorption peaks decreases evidently with the increasing DtFDA content, suggesting the addition of DtFDA leads to less ordered arrangement when compared with neat P(g2T-T) [46]. Moreover, ultrahigh-loading of the DtFDA (1:3 in mass) results in severe deviation of the absorption peak, leaving only one strong absorption peak at 563 nm. After thermal annealing (150 °C for 10 min on a hot plate in air), enhanced 0–0 vibronic transition near 662 nm is

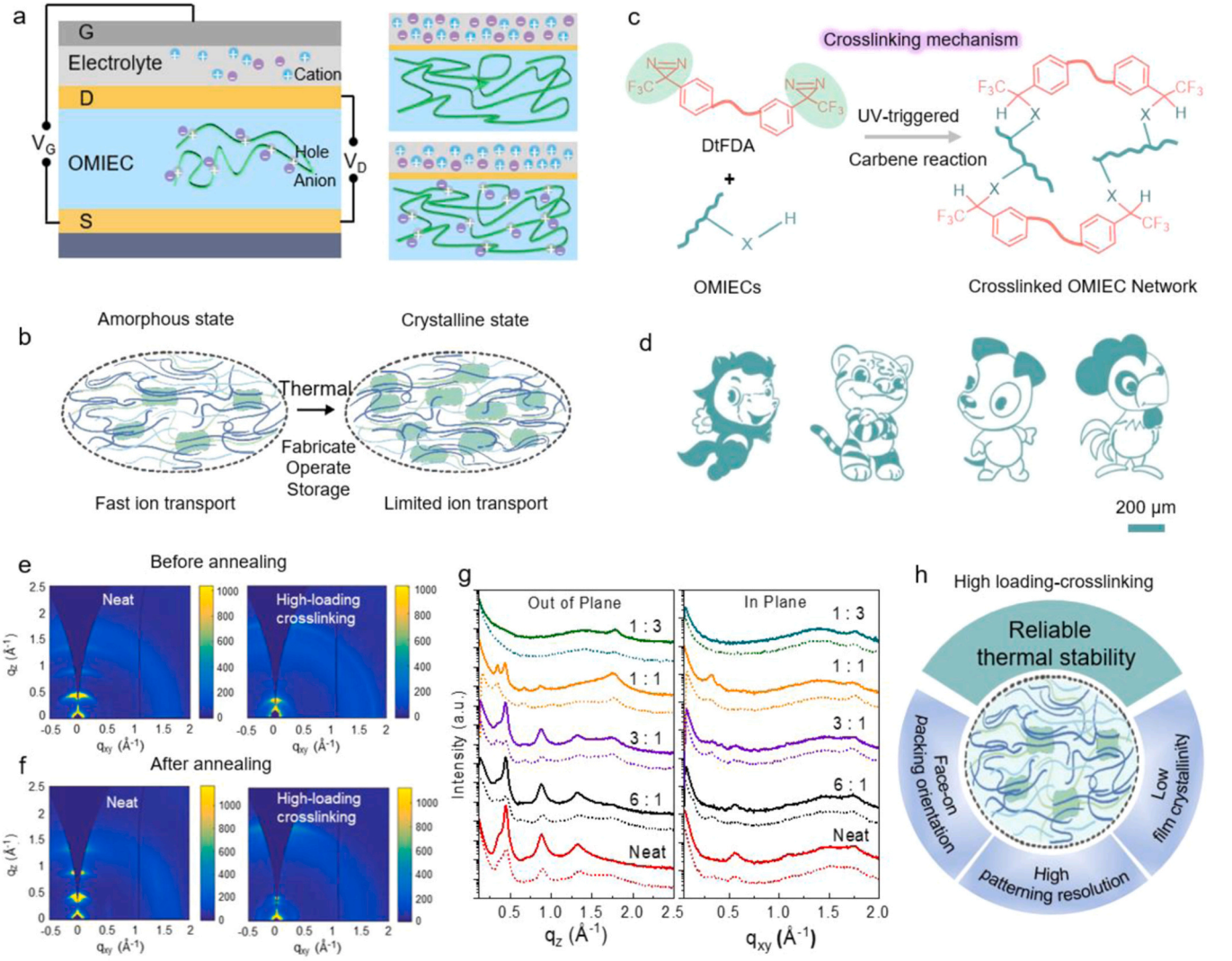


Fig. 1. Device structure, working principle, photopatterning mechanism, optical images, and microstructure of the P(g2T-T)/DtFDA blend films. a) Device structure and working principle of vOECT. b) Film instability due to thermal effects during fabrication, operation and storage of semiconductor films in vOECT. c) Direct photopatterning mechanism of the OMIEC/DtFDA blend film via UV-triggered carbene insertion crosslinking. d) Optical images of patterned P(g2T-T)/DtFDA blends, minimum line width for patterning is 10 μm. Two-dimensional GIWAXS patterns of e) unannealed and f) thermal annealed blend films. g) Out-of-plane (q_z) and in-plane (q_{xy}) line-cuts for P(g2T-T)/DtFDA blend films. Note, dash and solid curves corresponds to unannealed and annealed films, respectively. h) Advantages of high-loading crosslinking strategy.

detected for all the films, indicating an increase in the level of polymer aggregation after thermal stress. Note, for blend films with high DtFDA loading (1:1 and 1:3 in mass), no significant separated 0–0 and 0–1 peaks could be observed even after thermal annealing.

To further characterize the molecular packing variation due to the introduction of DtFDA and thermal stress, grazing incidence wide-angle X-ray scattering (GIWAXS) characterizations were conducted (Fig. 1**e-g** and Fig. S5). Neat P(g2T-T) film exhibit apparent h00 (100, 200 and 300) scattering peaks especially in out-of-plane direction, indicating high crystallinity. While gradually weakened crystallinity can be evidently observed, as out-of-plane 200 and 300 peaks constantly fade away along with higher DtFDA loading. This suggests that the DtFDA cross-linker in the blend films inhibits crystallization, which agrees with the results from AFM and UV–vis spectra. Besides, high-loading of the crosslinker altered the preferred polymer packing orientation from edge-on dominating to face-on dominating, as evidenced by the out-of-plane π-π* stacking diffractions (010) (Fig. 1**g**). Note that the (h00) peaks for all the films separate into double reflection families, which is attributed to the formation of two polymorphs or the presence of significant amounts of low molecular mass components due to the solubility difference of such polymer [47–50]. The separated dual reflection phenomenon in h00 peaks indicates wider d spacing can be formed, thus allowing sufficient permeation of solvated ions from electrolyte to move readily throughout the bulk of transistor channel [51]. Films subjected thermal stress were also characterized by GIWAXS (Fig. 1**f** and Fig. S5b). With relatively low DtFDA loading (P(g2T-T):DtFDA $\geq$ 3:1), intensities of the diffraction peaks are obviously enhanced after thermal stress; while with high DtFDA loading (P(g2T-T):DtFDA $\leq$ 1:1), no apparent variations on the diffraction patterns are observed. It is speculated that crosslinked polymer blend can lock the as-cast morphology and therefore conferring better thermal stability [45]. Moreover, after thermal stress, blend film with high-loading crosslinker (P(g2T-T):DtFDA mass ratio of 1:1) shows largely enhanced face on orientation, which should facilitate the charge transport along vertical direction. Therefore, the high-loading crosslinking approach is not only favorable for addressing the thermal stability bottleneck in vOECT, but also possesses the advantages of high patterning resolution, face-on packing orientation and low film crystallinity for mixed ionic-electronic coupling and transport

(Fig. 1**h**).

Next, vOECTs based on such blend films are fabricated and characterized (including both P(g2T-T)/DtFDA and PEG-N2200/DtFDA blend films with different mass ratios, detailed fabrication and characterization process can be found in Experimental and Fig. S6). Here, both source and drain electrodes hold a width of 30 μm (Note, drain electrode (top electrode) was thermal evaporated along with a cooling stage (5 °C) behind the substate holder, which can effectively minimize thermal annealing effect during thermal evaporation, *vide infra*), and the photopatterned channel blend films are 200 × 200 μm^2. Moreover, the vOECTs are encapsulated by Cin-Cell, where the channel area and source/drain electrodes pads for probe contacting are left open (Fig. 2**a**). Note that OECTs based on neat OMIEC films are not patterned and not encapsulated since they hold no cross-linking capability. Representative transfer, g_m, and output characteristics of the p-type P(g2T-T) based vOECTs are illustrated in Fig. 2**b** and Fig. S7. Introduction of the patterned P(g2T-T)/DtFDA channel layer and encapsulation would largely decrease the gate-current (I_G, dashed curves in Fig. 2**c**) from ~10^{-5} A in neat P(g2T-T) based vOECT to less than 10^{-6} A, along with slightly decreased off-current (I_{off}), which attribute to less parasitic capacitance. While negatively shifted turn-on voltage (V_{ON}) is observed along with higher DtFDA content (from 0.38±0.02 V in neat P(g2T-T) to 0.35±0.01, 0.33±0.02, 0.30±0.01, and 0.24±0.02 V in P(g2T-T)/DtFDA = 6:1, 3:1, 1:1, and 1:3, respectively). Similar phenomenon is also observed for the V_G that yields peak g_m (from 0.02±0.03 V in neat P(g2T-T) to −0.10±0.02, −0.12±0.02, −0.16±0.01, −0.33±0.02 V in P(g2T-T): DtFDA = 6:1, 3:1, 1:1, and 1:3, respectively). Therefore, increased DtFDA content in P(g2T-T) based vOECT would shift the overall transfer curve to more negative V_G region, due to less active semiconducting polymer and/or changed polymer packing in the channel.

On the other hand, peak g_m remain higher than 300 mS in these vOECTs based on P(g2T-T)/DtFDA blend channel (342.9±27.3, 333.5 ±20.4, 331.2±24.9, and 339.7±21.4 mS for neat P(g2T-T), P(g2T-T): DtFDA = 6:1, 3:1, and 1:1, respectively), except for vOECT with ultra-high DtFDA content (220.7±25.6 mS for P(g2T-T):DtFDA = 1:3). Especially, area normalized g_m of 378 μS/μm^2 belongs to the highest reported value to date. Moreover, the hysteresis in transfer characteristics reduces along with higher DtFDA loading (from 26.3±4.2 mV for neat P(g2T-T) to 22.5±2.7, 20.1±3.2, 4.9±1.3 and 6.8±1.9 mV for P(g2T-T):DtFDA = 6:1, 3:1, 1:1 and 1:3, respectively). Note, to quantify the hysteresis phenomenon, the voltage differences are obtained by calculating the halfway current between the on and off states ($|I_{DS}|$= 10^{-4} A for p-type and $|I_{DS}|$=10^{-6} A for n-type) [52,53], and the sweeping speed of V_G during the characterization is kept constant at 10 mV/s. Similar evolutions on the transfer and g_m characteristics are also observed in n-type OECT based on PEG-N2200 upon the addition of DtFDA (Fig. 2**c** and Fig. S8), where positively shifted transfer and g_m

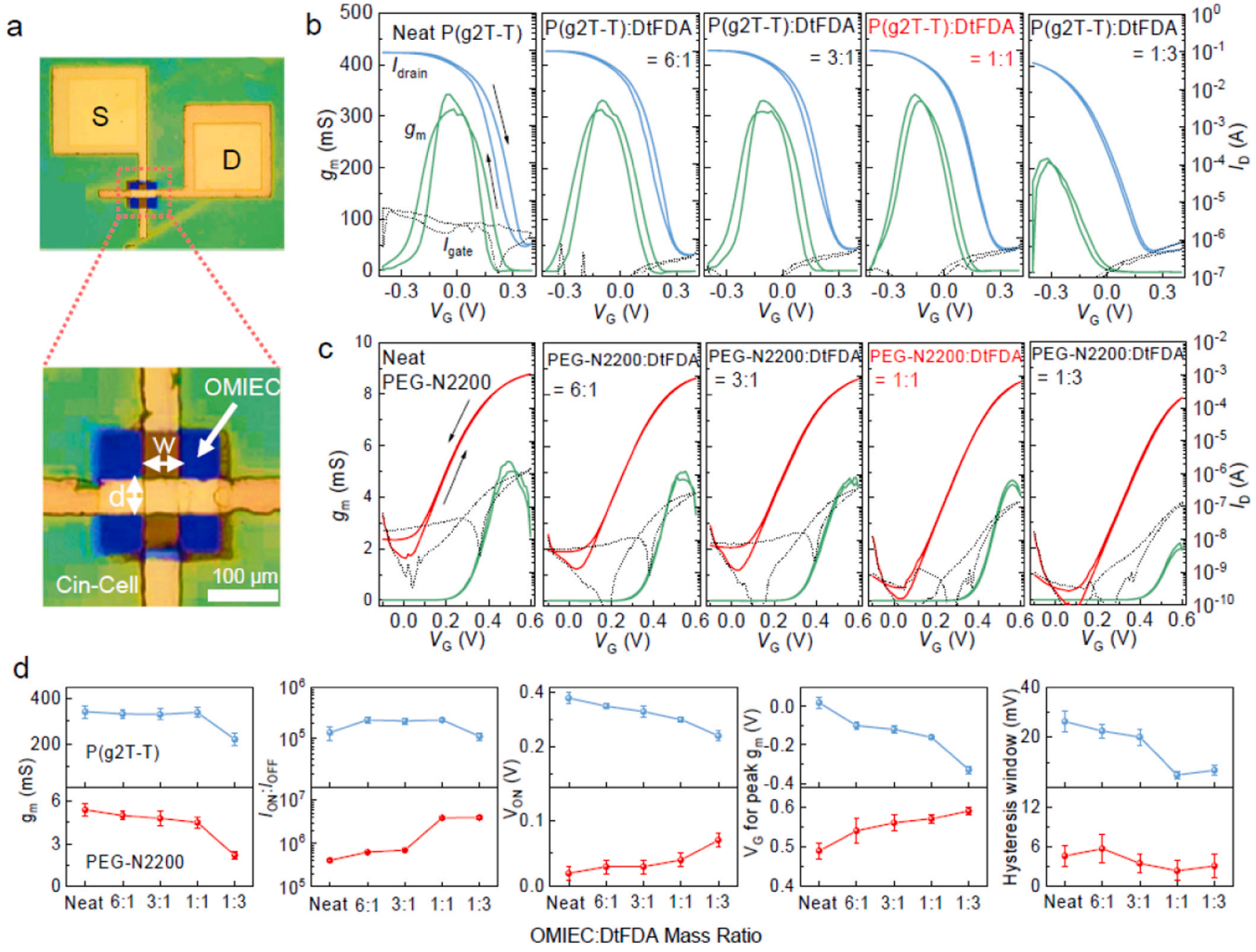


Fig. 2. vOECT structure and performance characteristics. a) Optical image of the fabricated vOECT. Transfer and transconductance characteristics of b) p-type and c) n-type vOECTs with different OMIEC:DtFDA mass ratio. V_D = −0.5 V and +0.5 V for p-type and n-type vOECTs, respectively. d) Correspondence of device parameters, including I_{ON}/I_{OFF}, V_{ON}, peak g_m, V_G yielding peak g_m and hysteresis window with different OMIEC:DtFDA mass ratio.

curves are observed with increasing DtFDA content. Detailed parameters extracted from transfer characteristics of both p- and n-type vOECTs are summarized in Fig. 2**d** and Table 1. It is obvious that DtFDA loading as high as 50 % of the total mass in the channel does not degenerate the overall performance of both types of OECTs. Moreover, in P(g2T-T) based vOECTs, addition of DtFDA leads to obviously decreased hysteresis, indicating possible enhancement on ionic transport.

Therefore, transient characteristics are next performed for both p- and n-type vOECTs (Figs. S9 and S10), and relevant transient time (τs) are further summarized in Fig. 3**a**. Turn-on transient time (τ_{ON}) of OECTs based on P(g2T-T) decreases more than 5 times, from 6.09±0.86 ms in neat P(g2T-T) to 3.76±0.61, 3.04±0.54, 1.88±0.32 and 1.04±0.18 ms in P(g2T-T):DTFDA = 6:1, 3:1, 1:1 and 1:3, respectively. Similar trend is also observed in turn-off transient time (τ_{OFF}), where it decreases from 9.58±1.46 ms in neat P(g2T-T) to 4.56±0.92, 0.73±0.23, 0.48±0.19 and 0.16±0.09 ms in P(g2T-T):DtFDA blends = 6:1, 3:1, 1:1 and 1:3, respectively. Note, such trend agrees well with the reduced crystallinity with higher DtFDA content, as amorphous OMIEC facilitates the diffusion of ions, thus leading to shorter τ. Same behavior is also observed in vOECTs based on PEG-N2200 (Fig. 3**a and** Fig. S10), where τ_{ON} decreases from 21.79±0.9 ms in neat PEG-N2200 to 5.55±0.62, 3.08 ±0.16, 1.92±0.28, and 1.31±0.22 ms in PEG-N2200:DtFDA blends = 6:1, 3:1, 1:1 and 1:3, respectively. While τ_{OFF} decreases from 3.99 ±0.50 ms in neat PEG-N2200 to 0.86±0.11, 0.70±0.06, 0.56±0.06, and 0.23±0.07 ms in PEG-N2200:DtFDA blend = 6:1, 3:1, 1:1 and 1:3, respectively. Note, the long τs in vOECTs based neat P(g2T-T) and PEG-N2200 are partially affected by the un-patterned channel and un-encapsulated electrodes, which would introduce large parasitic capacitance. While all the vOECTs based on OMIEC:DtFDA blend channels are patterned and encapsulated with the same process and geometry. Therefore, the shortened transient time along with higher DtFDA loading in these vOECTs is attributed to the weakened crystallinity of the channel blends.

Note, in the above vOECTs, to avoid thermal annealing of the OMIEC channel during the thermal evaporation of the top drain electrode, a cooling stage behind the substrate was applied (Fig. S11)[31]. Consequently, to evaluate the thermal adaptability of the OMIEC:DtFDA blend channel, vOECTs fabricated without cooling stage during thermal evaporation (Fig. S11b) are further characterized (Figs. S12 and S13). Similar trends on vOECT performances are observed as these in vOECT fabricated with a cooling stage, where adding DtFDA would not obviously affect g_m values (OMIEC:DtFDA < 1:3), but leads to overall negatively (for p-type) or positively (for n-type) shifted transfer curves (Fig. S14). Moreover, vOECT fabricated without cooling stage still show decreasing hysteresis window with increasing DtFDA loading, but the overall hysteresis window is larger than these in vOECTs fabricated with a cooling stage (Table 1 **and** Table S1). Such trend agrees well with the transient characteristics on these vOECTs (Fig. 3**a,** Figs. S15 and S16), where vOECTs based on either P(g2T-T) or PEG-N2200 all show longer τ_{ON}s and τ_{OFF}s when fabricated without a cooling stage, while adding more DtFDA shortens both τ_{ON}s and τ_{OFF}s. Especially, τ_{ON}s of vOECTs based on either neat P(g2T-T) or PEG-N2200 increase from 6.09±0.86 and 21.79±0.89 ms (with cooling stage) to 20.90±1.15 ms and 121.16 ±3.99 ms (without cooling stage), respectively, which are about 3- and 6-times increment. On the other hand, higher DtFDA loading not only significantly reduces τ, but also gradually minimizes the difference on τs between the vOECT fabricated with/without cooling stage. For instance, in vOECTs based on OMIEC:DtfDA =1:1, τ_{ON}s of p- and n-type vOECTs only increase from 1.88±0.32 and 2.12±0.28 ms to 2.93±0.41 and 4.28 ±0.21 ms, respectively. Detailed information on τs can be found in Fig. 3**a** and Table S2. Therefore, high DtFDA loading in the channel, would effectively improve the thermal adaptability, which not only can maintain high g_m, but also induce OMIEC channel with high ionic mobility during thermal evaporation.

In actual application scenarios, transistors need to hold high stability under different storage/working temperatures. Therefore, to further evaluate if the current vOECTs are compatible with variable ambient temperature fluctuations for operation and storage, thermal stability of these vOECTs (fabricated with a cooling stage) under continuous temperature regulations (from 40 to 100 °C) are also investigated. Note, here, vOECTs that have not been activated are defined as fresh ones (they have not been measured by applying an electrolyte and V_G, thus no ions were injected/ejected in the channel), while devices that have been activated are defined as un-fresh ones, meaning ions were injected/ejected in the channel. Representative transfer curves of fresh vOECT based on neat P(g2T-T) show significantly enlarged I_{OFF} and hysteresis window as the thermal annealing temperature increases, while such phenomenon is gradually suppressed via increasing DtFDA loading. As shown in Fig. 3**b** and Fig. S17a, almost identical transfer curves in p-type vOECT with high DtFDA loading (P(g2T-T):DtFDA ≤ 1:1) are obtained after annealing under various temperatures (40 ~ 100 °C). On the other hand, for un-fresh devices (Fig. 3**c** and Fig. S17b), transfer curve of vOECT based on neat P(g2T-T) show slightly decreased hysteresis window as the thermal annealing temperature increases, while a consistent low I_{OFF} is maintained after thermal annealing. Similarly, higher DtFDA loading also yields un-fresh vOECTs with better thermal adaptability. Compared to fresh vOECTs, un-fresh vOECTs show overall smaller variation on transfer characteristics after thermal annealing. Such difference is due to that the channels of un-fresh vOECTs were pre-activated by ion injection/ejection, which forms ion transport path in the channel. Consequently, along with residue ions and/or solvents in the channel, thermal stress would introduce less effect on the microstructure of transistor channel, leading to more stable transistor performance after thermal annealing. In addition, transient behaviors of these fresh and un-fresh vOECTs based on P(g2T-T), are also characterized. As shown in Fig. S18, τ_{OFF} of fresh vOECT based on neat P(g2T-T) are strongly influenced by thermal stress (Fig. 3**d**), which gradually increases from 11.1 ms (un-annealed) to 12.4, 13.8, 17.7 and 21.0 ms after 40 °C, 60 °C, 80 °C and 100 °C annealing, respectively, while τ_{ON} is less affected (Fig. S19). On the other hand, un-fresh vOECT based on neat P(g2T-T) reveals an opposite trend of τ_{OFF} after thermal annealing, where it decreases from 11.1 ms (un-annealed) to 9.9, 6.8, 5.0 and 3.6 ms after 40 °C, 60 °C, 80 °C and 100 °C annealing, respectively. It is speculated that such behavior is due to the pre-activated channel can

Table 1
Performance summary of both p- and n-type vOECTs with cooling stage during evaporation.

	Mass ratio	I_{ON} (A)	I_{ON}: I_{OFF}	V_{ON} (V)	V_G yields Peak g_m (V)	g_m (mS)	Hysteresis (mV)
P-type	Neat	$(9.5\pm0.5)\times10^{-2}$	$(1.3\pm0.4)\times10^{5}$	+0.38±0.02	+0.02±0.03	342.9±27.3	26.3±4.2
P(g2T-T):DtFDA	6:1	$(9.5\pm0.4)\times10^{-2}$	$(2.3\pm0.3)\times10^{5}$	+0.35±0.01	−0.10±0.02	333.5±20.4	22.5±2.7
	3:1	$(9.4\pm0.4)\times10^{-2}$	$(2.2\pm0.3)\times10^{5}$	+0.33±0.02	−0.12±0.02	331.2±24.9	20.1±3.2
	1:1	$(9.4\pm0.5)\times10^{-2}$	$(2.3\pm0.2)\times10^{5}$	+0.30±0.01	−0.16±0.01	339.7±21.4	4.9±1.3
	1:3	$(5.2\pm0.3)\times10^{-2}$	$(1.1\pm0.2)\times10^{5}$	+0.24±0.02	−0.33±0.02	220.7±25.6	6.8±1.9
N-type	Neat	$(1.1\pm0.1)\times10^{-3}$	$(4.1\pm0.2)\times10^{5}$	+0.02±0.01	+0.49±0.02	5.4±0.4	4.6±1.6
PEG-N2200:DtFDA	6:1	$(1.0\pm0.2)\times10^{-3}$	$(6.3\pm0.3)\times10^{5}$	+0.03±0.01	+0.54±0.03	5.0±0.3	5.7±2.1
	3:1	$(0.9\pm0.2)\times10^{-3}$	$(7.0\pm0.3)\times10^{5}$	+0.03±0.01	+0.56±0.02	4.8±0.5	3.5±1.4
	1:1	$(0.8\pm0.2)\times10^{-3}$	$(3.9\pm0.2)\times10^{6}$	+0.04±0.01	+0.57±0.01	4.5±0.4	2.4±1.5
	1:3	$(0.2\pm0.1)\times10^{-3}$	$(4.0\pm0.2)\times10^{6}$	+0.07±0.01	+0.59±0.01	2.2±0.3	3.1±1.7

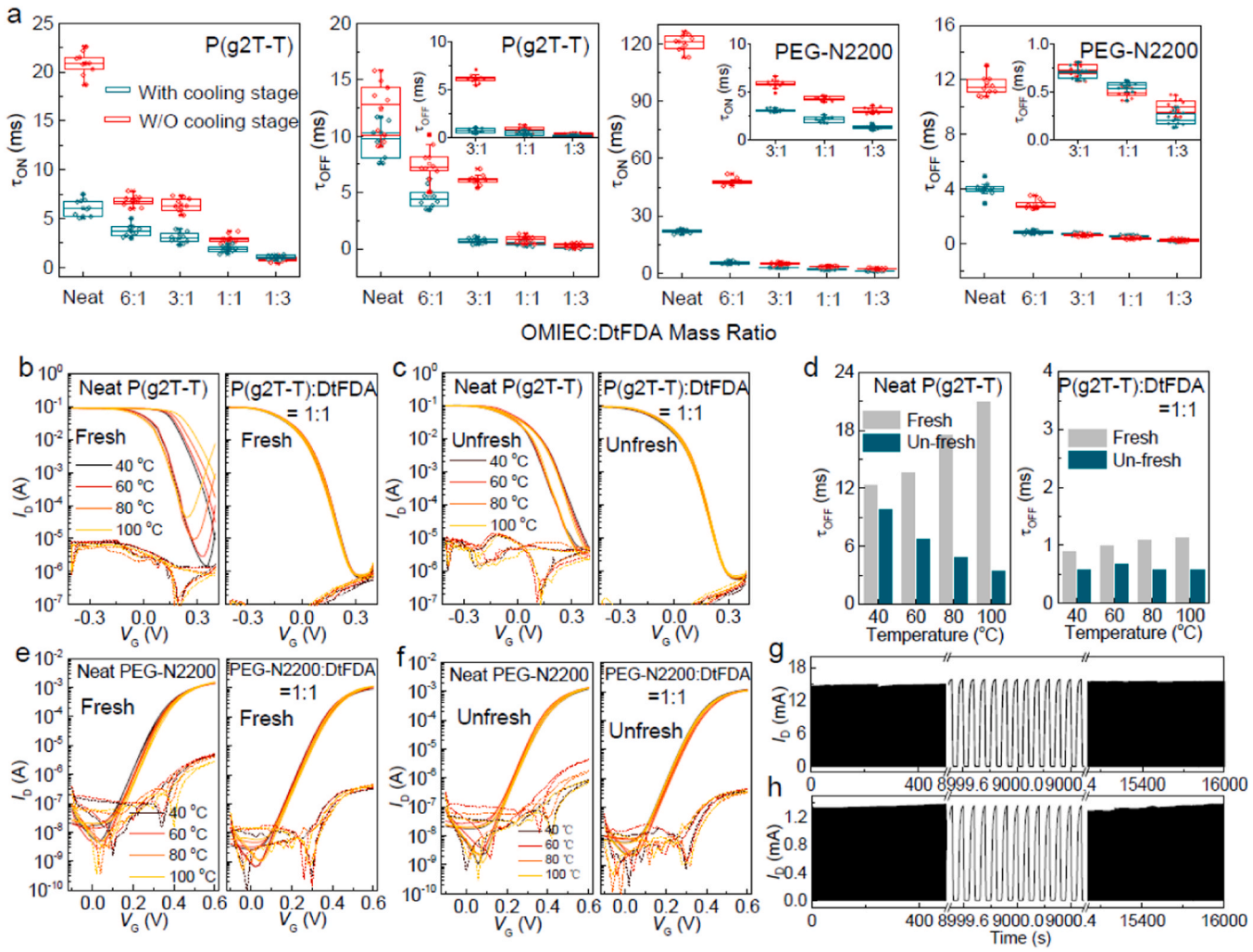


Fig. 3. Transient and stability characterizations of vOECTs based on OMIEC:DtFDA channel. a) Transient time of p- and n-type vOECTs fabricated with or without cooling stage, the error bars of transient time are calculated from 10 devices. Variations of transfer curves of b) fresh and c) unfresh p-type vOECTs after post thermal annealing. d) Comparison of τ_{OFF} in neat and high DtFDA loading (1:1 in mass) based p-type OECTs under thermal stress. Variation of transfer curves of e) fresh and f) unfresh n-type vOECTs after post thermal annealing. Cycling stability of high DtFDA loading (1:1 in mass) for g) p-type and h) n-type vOECTs.

maintain and facilitate ion transport even when annealed. It is worth noting that, both fresh and un-fresh vOECTs with high DtFDA loading (P(g2T-T):DtFDA=1:1) maintain fast τs ($\tau_{ON} \leq 2.3$ ms, $\tau_{OFF} \leq 1.1$ ms) after thermal stress.

Furthermore, by applying thermal stress to fresh vOECTs based on neat PEG-N2200, transfer curve is slightly positively shifted, indicating higher barrier for ion injection due to stronger crystallinity. While other vOECTs based on PEG-N2200:DtFDA blends or unfresh neat PEG-N2200 show minor variation on the transfer curves after thermal stress (Fig. 3**d-e** and Fig. S20). Transient characteristics on n-type vOECTs reveal similar trend as that in p-type vOECTs, where τ_{ON} and τ_{OFF} increases obviously as thermal annealing temperature increases for neat OMIEC based vOECTs, while vOECTs with high DtFDA loading (PEG-N2200:DtFDA = 1:1) maintain short τ_{ON} and τ_{OFF} of ≤ 1.9 ms and ≤ 0.7 ms, respectively (Figs. S21 and S22). These results strongly suggest that the high DtFDA loading in the vOECT channel can lock the amorphous morphology (*vide supra* GIWAXS), thus enabling better thermal stability to maintain efficient and stable ionic-electronic coupling and transport.

Next, cycling stability of the current vOECTs are assessed in atmosphere at room temperature, where a PDMS well is applied on the active channel area to hold ~ 5 μL electrolyte to ensure consistent measurement condition during cycling characteristics. As shown in Fig. S23, the switching characteristics of neat P(g2T-T) based vOECT was tested for 40,000 cycles (square wave of V_G between 0.4 V and −0.4 V under a frequency of 12.5 Hz, $V_D = -0.1$ V). When reaching to 39,000 cycles, the vOECT cannot be turned off effectively, which is also observed in the transfer curve (Fig. S24), where I_{OFF} of ~ 1 mA is obtained, which can be attributed to the damage of the transistor channel by material consumption and dissolution during continuous electrolyte inject/eject process, thus results in partially shorted source/drain electrodes of the ultrashort channel (~ 60 nm). On the other hand, vOECT based on P(g2T-T):DtFDA = 1:1 enables stable operation over 200,000 cycles (Fig. 3**g**), exhibiting negligible electrical performance degradation (Fig. S25a), which represents the highest record value of cycling stability for OECTs. With high DtFDA loading (PEG-N2200:DtFDA = 1:1), n-type vOECT also reveals extremely stable characteristic where more than 200,000 cycles with stable current switching and transfer curves are obtained (Fig. 3**h** and Fig. S25b). Moreover, it is important to note that due to the prevailing equipment testing conditions, the cyclic stability could have exceeded the currently reported values. These results indicate that the high DtFDA loading enables vOECTs with excellent cycling stability, which benefits from the highly crosslinked channel which that not only stabilize the channel microstructure, making it to be more resistant to swelling, and also secure the adhesion between the channel and top drain electrode. A thorough device performance comparison including cycling and thermal stability with recently published state-of-

the-art vOECTs are also summarized in Table S3, further demonstrating the breakthrough in electrical performance and stability as we have achieved with high-loading crosslinked approach.

The properties of ultra-stable, great signal amplification, and fast response speed make vOECTs suitable for implantable neural interfaces, which require reliable environmental adaptation and chronic tissue comfort, since these interfaces must operate reliably over prolonged tissue fluid immersion as well as storage under fluctuating brain temperatures, and any cycling and/or thermal instability may lead to device failure or degradation over time, resulting in ECoG signal distortion. As a proof of concept, ultraflexible vOECTs with biocompatible parylene substrates have been demonstrated for electrocorticography (ECoG) signal recording (Fig. 4**a**), with transistor channel of 30 μm for both source and drain widths and encapsulated with an exposed area of 200×200 μm^2. These vOECTs exhibit no obvious performance degradation, even after more than 500 deformations with bending radii of less than 1 mm, illustrating their considerable potential for use in soft, curved, and complex neural interfaces (Fig. S26). Through the craniotomy, the device was placed onto the rat's somatosensory area to receive electrical stimulation inputs from its hind limb (Fig. 4**b**). Note that the seamlessly attachment of the vOECT is crucial, as it connects directly to the cerebral cortex through a high-capacity, uniform crosslinking channel (Fig. 4**c**). For in vivo recording the brain activities via vOECT, the electrical activities from cerebral cortex act as gate inputs, and their changes induce variations in the channel current of OECTs. Somatosensory evoked potentials were successfully captured by administering external electric pulses at a frequency of 1 Hz and amplitude of 0.5 V to the hind limb of rats (Fig. 4**d**). According to the results of neural signals and associated power spectrum in Fig. 4**e**, such vOECT demonstrates a notable capability in the high-fidelity recordings of electrically stimulated neural signals and reveals the propagation of neural spikes and associated spatiotemporally resolved patterns. To evaluate the long-term reliability of such ultra-stable vOECTs, we also recorded the brain activities under the same electrically stimulation conditions after implantation for 4-weeks. Particularly, painkillers and anti-inflammatory drugs were injected into the rat for three consecutive days to suppress the pain and immune response brought about by craniotomy. The results also indicate that consistent ECoG recordings and power spectra were achieved, with no significant degradation in performance noted (Fig. 4**f**). Therefore, the high stability of the developed vOECT highlights the potential for implantable bio-applications where great thermal and cycling stability are needed.

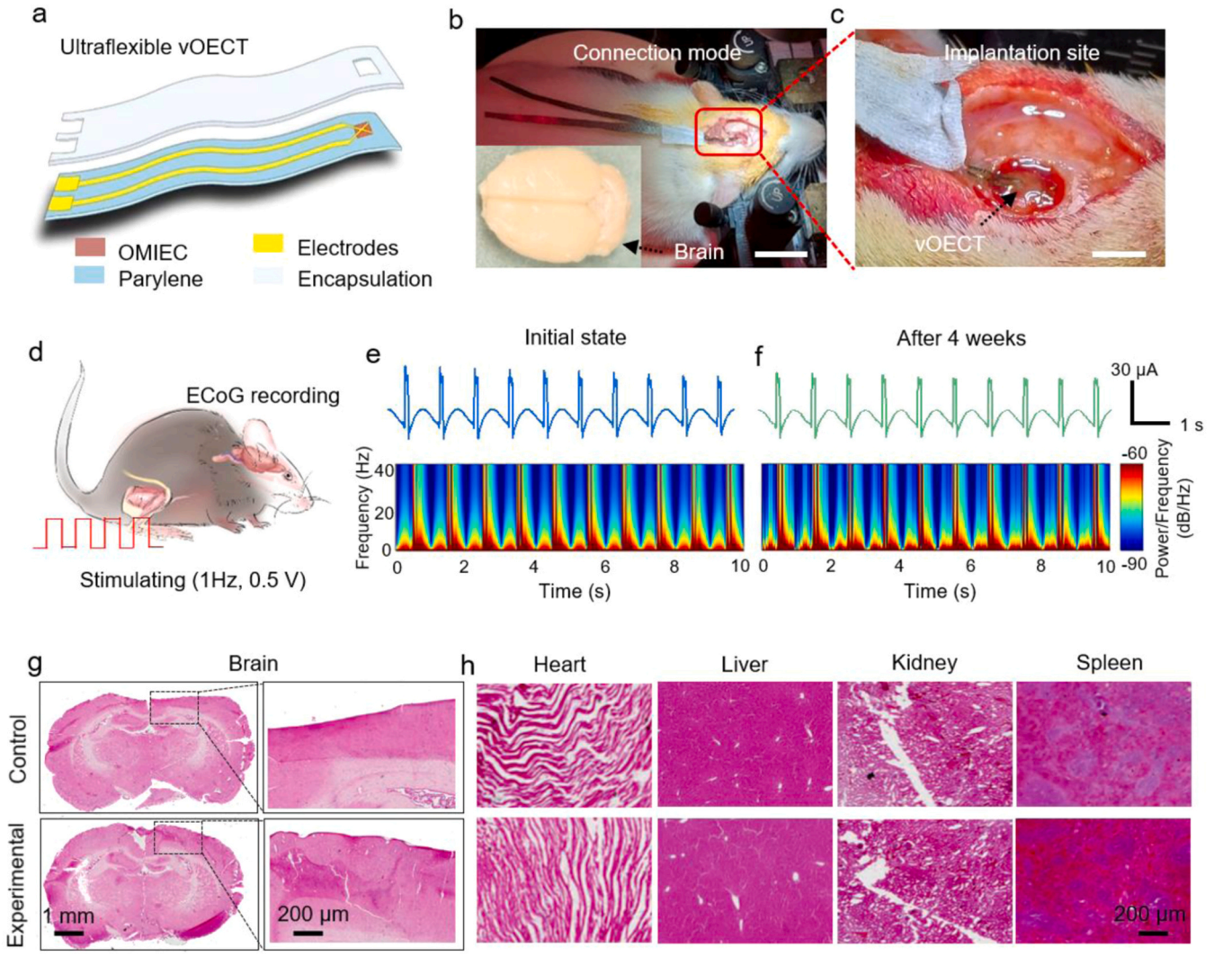


Fig. 4. Ultraflexible vOECT based implantable neural interface for stable ECoG signal recording with chronic biocompatibility. a) Illustration of the ultraflexible vOECT. b) Connection mode of implantable vOECT, inserted Fig. is the rat brain. Error bar, 2 cm. c) Location of implantable vOECT in regions of the rat cortex. Error bar, 8 mm. d) Schematic diagram for ECoG signal recording with stimulating approach, along with the recorded ECoG signals and corresponding spectrograms at the time of e) initial implantation and f) four weeks after implantation. g) Hematoxylin and eosin staining images of the Brain. h) Hematoxylin and eosin staining images of the Heart, Liver, Kidney and Spleen.

Furthermore, such vOECT demonstrates exceptional long-term biocompatibility, as confirmed by the Hematoxylin-Eosin staining results of tissue slices. In Fig. 4**g**, compared to the control group without vOECT implanting (top images), there is no generation of macrophages or astrocytes in the experimental group (bottom images). Besides, the H&E staining of viscera slices (heart, liver, kidney, and spleen) were performed to evaluate the biotoxicity of the long-term implanted vOECT. It turns out that in the aforementioned four tissues, both the control and experimental groups exhibit consistent cellular structure and composition, suggesting that the long-term implanted vOECT device did not cause any visceral toxicity (Fig. 4**h**). These results indicate that the rat tissues were not damaged during the surgery or over the whole implantation period. Consequently, such vOECT platform is expected to be a secure and biocompatible implantable neural interface, free from adverse reactions.

3. Conclusion

In this work, a highly loaded homogeneous crosslinking approach by adopting DtFDA crosslinker is proposed to derive ultra-stable vOECTs, in which OMIEC-based nanoscale morphologies variation and corresponding vOECT performance were carefully investigated in detail as proof of concept. It is found that the increased mass ratio of crosslinkers allows for efficient and fast ionic-electronic coupling and transporting in OECT channel, due to higher amorphous content and better resistance to thermal stress. Moreover, combining the excellent cycling and thermal stability with the intrinsic amplification capability of vOECT, continuous and stable ECoG signals could be readily monitored even after four weeks implantation. This work provides an effective and simple tool for the realization of vOECT channels with both high thermal and long cycling stabilities. We expect this design principle and our findings to be expanded into developing other organic electronics with excellent stabilities.

4. Experimental section

4.1. Materials

Semiconducting polymer P(g2T-T), PEG-N2200, cross-linker DtFDA and cinnamate-cellulose were synthesized according to previously reported procedures [31,40,42]. Dow Corning Sylgard 184 silicone elastomer clear kit was used to prepare PDMS well with base/crosslinker mix ratio of 10:1. Isopropyl alcohol and chloroform solvents were purchased from Sigma-Aldrich and used as received. Phosphate buffer (PBS, 1×) was provided by Solarbio Company and used as electrolyte. Ag/AgCl was used as gate electrode. To prepare relevant precursors for experiments, 15 mg/mL P(g2T-T) or PEG-N2200 and 20 mg/mL DtFDA solutions were first prepared with chloroform as solvent. Then, the P(g2T-T) or PEG-N2200 solution was mixed with DtFDA solution with a controlled volume ratio to obtain OMIEC/DtFDA blend solution with designated mass ratios. The blend solution was then stirred at room temperature for 2 hours. A solution containing cinnamate-cellulose with DtFDA crosslinker (10 mg/mL of cinnamate-cellulose and 2.5 mg/mL of DtFDA in chloroform) was prepared as the precursor of encapsulation layer.

4.2. Fabrication of vOECTs

For rigid vOECT, Si wafers (with 100 nm of SiO_2) as substrates were cleaned by ultrasonic with isopropyl alcohol for 15 min and dried in an oven (80 °C) for 8 hr. Then, 3 nm Cr and 150 nm Au were evaporated on the substrates as bottom source electrodes with a shadow mask. Next, the substrate with source electrodes were treated by ultraviolet ozone for 15 min, and transferred into a dry box (10 % RH, 25°C), where dense OECT channel films were deposited by spin-coating (3000 rpm, 30 s). The channel films were then photo patterned by exposure under 365 nm UV light (1500 mW) for 2 min with a shadow mask, and developed in chloroform for 1 s to remove un-crosslinked parts. Note, the thicknesses of all the channel films were ~60 nm. Then, 150 nm thick Au were evaporated (either with or without a cooling stage behind the substrates) on the channel layer as top drain electrodes with a shadow mask. Next, cinnamate-cellulose with DtFDA were subsequently spin-coated on top of fabricated devices at 5000 rpm for 30 s, followed by photopatterning (365 nm UV light for 2 min) and developing (chloroform) process.

For ultraflexible vOECT, a layer of micro-90 (Cole-Parmer) was spin-coated (3000 rpm, 60 s) on Si wafers as sacrificial layer, to obtain silicon/micro-90 support. Subsequently, a 1-μm-thick parylene film was deposited onto the silicon/micro-90 substrate utilizing parylene coater (Specialty Coating System, PDS 2010, KISCO Company) to serve as substrate. Prior to the deposition of the bottom source electrode, the silicon /micro-90 / parylene support underwent plasma cleaning for 5 min. And followed by the same fabrication producer with rigid vOECT. Finally, to obtain ultraflexible vOECT, the edges of the device were gently touched with deionized water to facilitate readily peeling.

4.3. Film characterization

Patterned OMIEC/DtFDA patterns with 1 μm resolution by direct photolithography were characterized by optical microscope. Atomic force microscope (AFM) images were characterized by Bruker Dimension Icon in tapping mode. UV–vis absorption spectra were measured using SHIMADZU UV-1700 with the films deposited on glass substrate. GIWAXS measurements were recorded at BeamLine 8-ID-E1 at the Advanced Photon Source (APS) at Argonne National Laboratory.

4.4. In Vivo measurement

All procedures were conducted in accordance with the ethical guidelines of the Department of Health of Kong Hong, animal studies and the experimental protocol was reviewed and approved by the Institutional Animal Care and Use Committee at the City University of Hong Kong (approval number A-0664). The procedure commenced by anesthetizing the heads of 4–6 weeks old rat using a stereotaxic apparatus, achieved through intraperitoneal injection of ketamine. Once the rat reached a state of deep anesthesia (~10–15 minutes), the scalp was incised using sterilized surgical scissors to expose the skull. An anesthesia mixture of isoflurane and oxygen was swiftly administered and withdrawn through a tracheal tube inserted into the rat's mouth. Throughout the experiment, real-time monitoring of heart rate and blood pressure was conducted to ensure the maintenance of anesthesia depth and stability. In addition, a heating blanket was employed to sustain the rat's body temperature at 37°C. Craniotomies were performed using a handheld electric drill equipped with a craniotomy bit approximately 0.5 mm in diameter. Following the removal of the dura mater, an ultraflexible vOECT coated with gel foam and saline was positioned over the exposed cerebral cortex, and both the skull and OECT were then covered with dental cement. Furthermore, the connecting plug was securely fastened to the cement. To alleviate postoperative discomfort, the rats received daily doses of meloxicam. Following a three-day recovery period, neural signals were recorded from the rat.

4.5. Chronic evaluation of immunohistochemistry

Rats were anesthetized and exposed to carbon monoxide (CO) at concentrations ranging from 0.2 % to 0.5 %. Following exposure, rat brains were excised and subsequently immersed in a 4 % formaldehyde solution at 4°C overnight, prior to cryopreservation in a 30 % sucrose (SU) solution for a duration of 3 days. Coronal sections of 20 μm thickness were then obtained utilizing a cryostat and affixed onto replacement slides maintained at −20°C until subsequent processing.

For immunostaining, a series of preparatory steps were undertaken. Initially, the slides were rinsed with distilled water, followed by incubation with hematoxylin stain for 5 minutes at room temperature. Next, the slides were immersed in a solvent mixture comprising 0.1 % hydrochloric acid and ethanol for a duration of 1–3 s, followed by rinsing with distilled water for one minute to facilitate differentiation. The slides were subsequently immersed in a 1× PBS solution for 30 s, followed by a 3 minutes rinse with distilled water. The slides were then rinsed in 95 % ethanol for 5–10 s and counterstained with eosin stain for 30 s. Finally, the slides were dehydrated twice in ethanol for 3 minutes each and washed twice in dimethylanisole for 3 minutes each. All these slides were observed on an optical microscope and images were captured.

4.6. Transistor characterization and ECoG recording

Electrical properties of all the vOECTs were performed using a semiconductor analyzer (FS-Pro, PDA) with a probe station (Chengdu ChipTest Technology Co., LTD.). For ECoG signal recording, the data were acquired by a PowerLab data collector (PL3516/P PowerLab 16/35, AD Instruments) with a sampling frequency of 10,000 Hz. Next, ECoG data were digitally bandpass filtered (1–40 Hz) by matched software Origin and then processed with a customized program in MATLABR2022a to obtain power spectrum.

CRediT authorship contribution statement

Yujie Peng: Validation, Formal analysis. **Mengge Wu:** Visualization, Methodology, Investigation. **Libing Bai:** Resources, Data curation. **Miao Xie:** Methodology, Formal analysis. **Yueping Lai:** Validation, Resources, Methodology, Formal analysis. **Dayong Zhang:** Resources, Investigation. **Changjian Liu:** Validation, Software. **Wei Huang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Liang-Wen Feng:** Writing – review & editing, Resources, Methodology. **Xinge Yu:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Qiang Zhang:** Methodology, Investigation, Data curation. **Junsheng Yu:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Lin Gao:** Writing – original draft, Validation, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nanoen.2024.110062.

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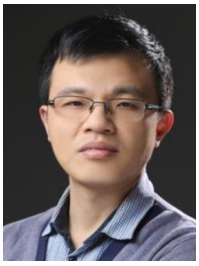
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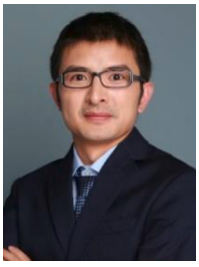
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